Direct Multi-Horizon GATv2 for Train Delay Prediction

Thinh Duy Do
Halicioglu Data Science Institute
University of California, San Diego
San Diego, CA, USA

***Abstract*—Train-delay forecasting is challenging because railway operations contain strong spatial and temporal dependencies among trains, stations, and operational events. Recent graph-based approaches use railway topology to model these relationships, but multi-step autoregressive forecasting may introduce error accumulation because earlier predictions are recursively reused in later forecasting steps. This work investigates a direct multi-horizon Graph Attention Network v2 (GATv2) framework for train-delay prediction. The proposed approach retains an event-based graph representation, a GATv2 encoder, and a two-hurdle classification–regression architecture, while replacing autoregressive decoding with direct prediction of multiple future horizons. The study will compare the proposed architecture with persistence and autoregressive GATv2 baselines using both regression and classification metrics. The central objective is to determine whether direct multi-horizon forecasting can reduce horizon-dependent prediction error while preserving the benefits of topology-aware railway-delay modeling.**

***Index Terms*—train delay prediction, graph neural networks, GATv2, multi-horizon forecasting, delay propagation**

I. Introduction

Public transit and railway systems support large-scale passenger mobility and play an important role in transportation infrastructure. Service disruptions can affect passengers, operators, and the wider transportation network, making reliable train operations an important operational objective. In the United States, public transportation systems support billions of passenger trips each year, illustrating the scale at which transportation reliability can affect daily mobility. Train delays may also impose economic and operational costs when disruptions interfere with passenger travel, scheduling, and network capacity. These impacts motivate the development of methods capable of anticipating delay conditions before they spread through the railway system.

Railway operations consist of deeply connected train events rather than isolated observations. An initial disturbance, referred to as a *primary delay*, can influence nearby and subsequent train events through operational dependencies such as shared infrastructure, dwell interactions, and headway constraints. These interactions may generate *secondary* or *knock-on delays*. The process through which delays spread across successive train events is known as *delay propagation*, and repeated propagation can produce cascading effects across the railway network.

Consequently, railway-delay evolution contains both spatial and temporal dependencies. The state of one train event may depend not only on its own previous state but also on structurally related neighboring events. Graph-based learning provides a natural framework for representing these relationships by modeling train events as nodes and operational dependencies as edges. Its learning provides a natural connection between these paradigms by representing train-event dependencies explicitly through graph structure while learning nonlinear delay propagation patterns from operational data [1]. Through graph message passing, information from connected railway events can contribute to the representation and prediction of a target event.

Recent graph-based approaches have demonstrated the potential of topology-aware learning for railway-delay prediction. Nguyen and Li [2] proposed an event-based GATv2 framework in which train events are represented as graph nodes and operational relationships, including running, dwell, and headway interactions, are represented as edges. Their framework combines delay classification with delay-magnitude regression and generates multi-step forecasts using an autoregressive rollout strategy.

Although autoregressive forecasting provides a natural representation of sequential evolution, it recursively reuses predicted states when producing later forecasts. As a result, prediction errors generated at an earlier step may become inputs to subsequent forecasting steps and potentially contribute to increasing error as the forecasting horizon becomes more distant. This creates an unresolved question: whether multi-step forecasting degradation is primarily associated with the graph representation itself or with the autoregressive decoding strategy used on top of that representation.

This study retains the event-based data representation, graph construction, node and edge features, GATv2 encoder, and two-hurdle classification–regression framework while replacing autoregressive decoding with direct multi-horizon prediction. Instead of recursively feeding a predicted delay into the next forecasting step, the proposed model predicts multiple future horizons directly from the current graph representation.

The study investigates a main research questions whether direct multi-horizon decoding can reduce horizon-dependent error accumulation relative to autoregressive forecasting and thereby improve delay-magnitude prediction, particularly at more distant forecasting horizons. The intended contributions of this work is a Direct Mutlti-Horizon GATv2 framework for the train delay prediction that forecasts a sequence of

train events and evaluate if the model can reduce the error accumulation at longer horizons.

II. Related Work

A. Graph Convolutional Network (GCN) for Train Delay Prediction

Graph Convolutional Networks (GCNs) are designed to learn from graph-structured data by combining node features with graph connectivity [3]. Through graph convolution, each node aggregates information from its neighboring nodes, allowing the model to incorporate relationships between connected entities. GCNs provide a simple and computationally efficient framework and have become a foundational architecture for modern message-passing graph neural networks.

Although GCNs are effective at combining information from a node and its neighbors, the standard GCN formulation does not explicitly learn the relative importance of individual neighboring nodes. Instead, neighboring information is aggregated using normalized graph connectivity. This limitation may reduce the model's ability to distinguish which neighboring events contribute more strongly to train delay propagation. This motivates the use of Graph Attention Networks (GATs), which learn different attention weights for neighboring nodes.

B. Autoregressive Multi-Step Forecasting

Autoregressive multi-step forecasting generates future predictions sequentially, where an earlier prediction is reused when generating a later prediction [2]. The model first predicts the next event and then recursively incorporates predicted information into subsequent forecasting steps. This approach naturally represents the sequential structure of railway operations, where train events are dependent on previous and neighboring events.

Nguyen and Li [2] introduced an event-based GATv2 framework for multi-step train delay prediction using an autoregressive rollout. Their model represents train events as graph nodes and models operational relationships between events, allowing the graph structure to capture how delays propagate through the railway network. Their framework demonstrates the benefit of combining railway topology with graph-based learning for identifying and estimating train delays.

However, autoregressive forecasting is susceptible to error accumulation because predictions generated at earlier horizons are reused when predicting later events. An inaccurate prediction at one step may therefore influence subsequent predictions, causing prediction errors to propagate as the forecasting horizon increases. In addition, train delays may remain relatively stable across consecutive events, making a persistence baseline difficult to outperform in some forecasting conditions.

The framework proposed by Nguyen and Li [2] serves as the primary methodological basis for this study. Building upon their event-based graph representation and GATv2 architecture, this work investigates whether replacing autoregressive forecasting with direct multi-horizon prediction can reduce horizon-dependent error accumulation. Instead of recursively feeding previous predictions into later forecasting steps, the proposed approach directly predicts multiple future horizons from the current graph representation.

III. Methodology

A. Problem Formulation

The train delay prediction problem is formulated over an event-based railway graph

$$G = (V, E), \tag{1}$$

where V denotes the set of train events and E represents operational dependencies between events. Each node $v_i \in V$ corresponds to a train event and is associated with a feature vector

$$\mathbf{x}_i \in \mathbb{R}^F, \tag{2}$$

where F is the number of input features. Edges represent operational relationships such as running (a train departing from previous station and arriving at the next station), dwelling (a train arriving at the current station and departing from it), and headway interactions (the time between two consecutive trains) between train events.

For each event v_i, let y_i denote its observed delay in seconds. Given the graph structure and node features available at the current event, the objective is to estimate the delays of the next K future events. In this study, the forecasting horizon is fixed to $K = 3$. The prediction target for event v_i is therefore defined as

$$\mathbf{y}_i = \left[y_i^{(1)}, y_i^{(2)}, y_i^{(3)} \right], \tag{3}$$

where $y_i^{(k)}$ represents the observed delay at the k-th future event relative to the current prediction point.

The autoregressive baseline generates these predictions sequentially. A prediction produced at an earlier horizon is reused when estimating a later horizon, which can be represented conceptually as

$$\hat{y}_i^{(1)} \rightarrow \hat{y}_i^{(2)} \rightarrow \hat{y}_i^{(3)}. \tag{4}$$

As a result, prediction errors generated at earlier horizons may propagate into subsequent forecasts.

In contrast, the proposed Direct Multi-Horizon (DMH) formulation predicts all three horizons directly from the learned graph representation:

$$\hat{\mathbf{y}}_i = \left[\hat{y}_i^{(1)}, \hat{y}_i^{(2)}, \hat{y}_i^{(3)} \right]. \tag{5}$$

Thus, later-horizon predictions do not require previously predicted delay values as recursive inputs. The primary objective of this study is to compare the autoregressive and direct multi-horizon forecasting strategies under a common event-based graph representation and GATv2 architecture.

B. Data and Preprocessing

The dataset is downloaded from Subway Data NYC including New York City subway observations [4]. It is attached by the official MTA static GTFS schedule. Each data provides different type of information such as the observed data provides train trips, stops, and arrival/ departure times, whereas the static GTFS data provides scheduled trips and event times. Moreover, observed trips link to the static GTFS schedules by matching service day.

C. Feature Engineering

Feature engineering is essential to capture the interrelationship between nodes. Three main operational features were derived. Dwell time represents the time a train remains at a station:

$$D_i = t_i^{dep} - t_i^{arr}. \tag{6}$$

Running time represents the travel time between two consecutive stops:

$$R_i = t_i^{arr} - t_{i-1}^{dep}. \tag{7}$$

Headway represents the temporal separation between consecutive trains serving the same route, direction, and stop:

$$H_i = t_i - t_{i-1}. \tag{8}$$

Trip context was represented by stop sequence, normalized trip progress, and the number of stops remaining. Time of day was encoded using sine and cosine transformations together with a rush-hour indicator.

D. Graph Construction

The completed graph is built based on gathering all engineered and collected features. For the node, each of arrival and departure is considered as a solid node (event-based). The event nodes are connected by three engineered edge types. A Dwell edge connects an arrival to its corresponding departure at the same stop, while a Run edge connects a departure to the subsequent arrival of the same train. A Headway edge connects consecutive train events occurring on the same route, direction, stop, and event type.

E. GATv2 Backbone and Two-Stage Hurdle Model

When the complete graph is formed, the node feature matrix is first passed through a three-layer GATv2 backbone [5]. Each layer aggregates information from neighboring train events using learned attention weights, allowing the model to capture delay propagation through the relationship of dwell, headway, and running. The output of the final GATv2 layer is a latent representation for each event node.

The result continues to put through in a two-stage hurdle model. The first stage is a binary classifer where the model filtering an event that actually is delayed. The second-stage regressor only takes place when the event is predicted as delyed. The final prediction is therefore

$$\hat{y} = \begin{cases} \hat{y}_{reg}, & P(\text{delay}) \geq 0.5, \\ 0, & \text{otherwise.} \end{cases} \tag{9}$$

This separates the prediction of delay occurrence from the prediction of delay severity.

F. Proposed Direct Multi-Horizon Forecasting

The proposed Direct Multi-Horizon (DMH) model extends the GATv2 framework to predict the next three operational events simultaneously. DMH tend to predict three future horizons from the current information's node, it would avoid the error from the previous horizon continues to increase at the next prediction. For each source event i, the GATv2 representation is passed to three-horizon classifier and regression heads, producing

$$[\hat{y}_{i,t+1}, \hat{y}_{i,t+2}, \hat{y}_{i,t+3}] = f_\theta(\mathbf{X}, E). \tag{10}$$

Here, $t{+}1$, $t{+}2$, and $t{+}3$ correspond to the next three arrival or departure events within the same trip. Unlike autoregressive forecasting. Because a future event may be predicted by several earlier source events at different horizons, with the same target, it would contains three different prediction at different k-ahead. For horizon k, the weight formula is defined as: event. These estimates are combined using validation-based horizon weights. For horizon k, the weight is defined as

$$w_k = \frac{1/\text{MAE}_k}{\sum_{r=1}^{3} 1/\text{MAE}_r}, \tag{11}$$

where MAE_k is the validation MAE of horizon k. The final delay estimate for event j is

$$\hat{y}_j = \frac{\sum_{k \in \mathcal{A}_j} w_k \hat{y}_j^{(k)}}{\sum_{k \in \mathcal{A}_j} w_k}, \tag{12}$$

where $\mathcal{A}_j$ contains the available predictions for event j. This gives greater influence to horizons that demonstrate lower validation error.

IV. Experimental Setup

A. Baselines

The planned experimental comparison based on regression metrics such as MAE, and RMSE and categorical metics such as Accuracy, Precision, Recall and F1-Score including three principal forecasting approaches:

- Zero baseline
- Persistence baseline.
- Autoregressive GATv2.
- Proposed direct multi-horizon GATv2.

V. Results and Discussion

A. Overall Forecasting Performance

Model	MAE	RMSE	Accuracy	Precision	Recall	F1-Score
Zero-Delay	6.7103	13.7107	0.5167	0.0000	0.0000	0.0000
Persistence	**0.7290**	**1.8926**	0.9290	0.9382	0.9132	0.9255
Autoregressive GATv2	3.1243	7.5966	0.9223	0.9380	0.8987	0.9179
DMH GATv2 + Aggregation	2.5599	6.9135	**0.9458**	**0.9494**	**0.9378**	**0.9436**

TABLE I
Overall Validation Performance of the Four Models

Overall, the persistence baseline shows its strength and effective in train delay prediction since the delay rarely observe big changes. It allows the model to avoid previous error prediction influence the next one compared to GATv2. Regarding to the classification aspect, the GATv2 is outstanding and show its reliability in playing a major role in delay detection.

This study successfully keeps the strength of autoregressive in topology exploration and error accumulation improvement. Appararently, direct multi-horizon outperform the other in delay detection evaluation metrics. Specifically, DMH detects correctly about 94.58% compared to 93.8% whereas DMH also capture more than 4% in delay recognition (Recall) at 93.78% . Even though the persistence baseline is hard to overcome in MAE measure, the paper still keep the importance in decision support tools.

B. Horizon Forecasting Performance

Model	MAE (min)			Accuracy		
	$t+1$	$t+2$	$t+3$	$t+1$	$t+2$	$t+3$
Autoregressive GATv2	**2.411**	2.861	3.124	**0.9570**	0.9331	0.9221
DMH GATv2	2.645	**2.703**	**2.751**	0.9568	**0.9362**	**0.9258**

TABLE II
Horizon-Wise Comparison of Autoregressive and Direct Multi-Horizon Forecasting

In term of MAE, the autoregressive GATv2 performes slightly better than direct multi-horizon GATv2 in MAE(min) at $t+1$. As the prediction moves forward to $t+2$ and $t+3$, performance gap becomes more noticeable when DMH predicts more precise at 2.751 minutues compared to 3.124 of the other model. With regard to Accuracy, the same trend occurs when DMH slightly overcomes Autoregressive as later horizon. The table shows the strength of this study in stable delay prediction, which is a crucial characteristic train operators looking for at a model.

VI. Limitations and Future Work

The current study uses a fixed number of forecasting horizons, which illustrate the short-range of delay prediction. The change in used quantity possibly changes the performance of the model. Future work could investigate that current state and try multiple settings to identify the optimal horizon length.

The stability in performance of Persistence model emphasizes the characteristics of train system when there is rarely a big jump takes place in the sequence. Therefore, in some cases, the learned models might not be optimal for correctly predicting. However, the topology's strength is valuable for delay detection as a supported tool for train operators. Future work should test more on a variety of train system that there are more occurrance of sudden disruptions and congestion

VII. Conclusion

The study investigates error accumulation for GATv2 using a short horizon length. Perform the comparison between Autoregressive and DMH observer an innovative step in strengthening the topology for train delay forecasting. The DMH model performs slightly better and is more stable compared to the other. Moreover, the study also proves that the prediction of multiple future train events directly from the current graph state with three horizons, helping the model perform better at MAE evaluation metrics. It is a promising study that machine learning could benefit from topology and learn the delay more precisely in the future work depending on analyzing DMH on a variety numbers of horizons.

References

[1] T. Spanninger, A. Trivella, B. Büchel, and F. Corman, "A review of train delay prediction approaches," *Journal of Rail Transport Planning & Management*, vol. 22, Art. no. 100312, 2022, doi: 10.1016/j.jrtpm.2022.100312.

[2] V. D. A. Nguyen and Z. Li, "Deep Learning to Identify the Spatio-Temporal Cascading Effects of Train Delays in a High-Density Network," in *Proc. 8th ACM SIGSPATIAL Int. Workshop on AI for Geographic Knowledge Discovery (GeoAI '25)*, Minneapolis, MN, USA, 2025, pp. 1–11, doi: 10.1145/3764912.3770828.

[3] T. N. Kipf and M. Welling, "Semi-Supervised Classification with Graph Convolutional Networks," in *Proc. International Conference on Learning Representations (ICLR)*, 2017.

[4] "Subway Data NYC," [Online]. Available: https://subwaydata.nyc/. [Accessed: Sep. 1, 2026].

[5] S. Brody, U. Alon, and E. Yahav, "How Attentive Are Graph Attention Networks?" in *Proc. International Conference on Learning Representations (ICLR)*, 2022.